

Solution Architecture

Date	04 July 2026
Team ID	SOLO-CCFD-2026
Project Name	Credit Card Fraud Detection using Machine Learning
Maximum Marks	5 Marks

High-level architecture of the Credit Card Fraud Detection solution across presentation, application, and data layers.

Presentation / Client Layer
Web App (Flask + HTML/Bootstrap) — Analyst Input Form & Results View

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API Gateway / Application Layer
Flask REST routes (/predict, /batch-predict)

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Core ML Scoring Service External Integration
(Preprocessing + best trained IBM Watson Machine Learning
model: XGBoost/RF/LogReg/DT) (hosted deployment endpoint)

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Data / Storage Layer
creditcard.csv (training data), trained_model.pkl (model store),
prediction_logs (optional SQLite)

Component Description Table

Component	Description / Role	Technologies Used
Presentation Layer	Form for entering/uploading transaction features and viewing fraud/genuine result	HTML, CSS, Bootstrap, Jinja2
Application / API Layer	Receives requests, preprocesses features, calls the ML model, returns prediction	Flask, Python
Core Logic Service	Loads trained classifier and scaler, computes fraud probability	scikit-learn, XGBoost, joblib/pickle
External Integration	Hosted scoring endpoint for scalable real-time predictions	IBM Watson Machine Learning REST API
Data / Storage Layer	Stores raw dataset, trained model artifact, and optional prediction logs	CSV, Pickle, SQLite (optional)